

## Mollic Andosols

### Description:

Mollic Andosols are **volcanic soils** formed from **volcanic ash, pumice, or other pyroclastic materials**. They possess a **dark, humus-rich surface horizon (mollic horizon)** and are characterized by excellent structure, high porosity, and strong aggregation.

These soils are typically light and friable, allowing for easy cultivation and good aeration. Their formation occurs under **cool, humid conditions** that favor the accumulation of organic matter and the weathering of volcanic glass into clay minerals such as allophane and imogolite.

### Drainage:

- **Wet season:** Generally well-drained, with rapid water infiltration due to high porosity. However, in depressions or poorly drained areas, temporary waterlogging can occur.
- **Dry season:** Retain adequate moisture longer than many soils due to high organic matter and fine pore structure, but can dry out under prolonged drought.

### Landscape:

Commonly found in **volcanic highlands** and **upland plateaus**, often on gently to moderately sloping terrain formed by volcanic activity.

### Depth:

Usually moderately deep to deep, depending on the thickness of volcanic ash deposits. Rooting depth is good, allowing plants to explore nutrients effectively, though restricted in areas with rock fragments or compact sublayers.

### Workability:

- **When wet:** Soft and friable, allowing for easy tillage but may become sticky if too moist.
- **When dry:** Light and crumbly, easy to cultivate and ideal for seedbed preparation.

Overall, Mollic Andosols have **excellent tilth and workability**, making them among the best agricultural soils.

### **Nutrient Richness & Cation Exchange Capacity (CEC):**

- **CEC:** Moderate to high, due to high organic matter and amorphous clay minerals.
- **Base saturation:** Usually moderate; cations like  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$ , and  $\text{K}^+$  are present but can be depleted over time.
- **Organic matter:** Very high, contributing to dark color and excellent nutrient-holding capacity.
- **Nutrients:** Despite good fertility, **phosphorus fixation** is a key limitation. P becomes bound to allophane and iron/aluminum oxides.
- **Nitrogen:** Often sufficient due to high organic matter but may decline with continuous cropping.

### **pH:**

Generally **slightly acidic to neutral (5.5–7.0)**, depending on rainfall, vegetation, and parent material. Liming may be needed in highly leached areas.

### **Management:**

- Apply **phosphorus fertilizers** in available forms (DAP, TSP, or rock phosphate).
- Lime periodically to counteract acidity and maintain optimal nutrient availability.
- Incorporate organic matter (manure, compost, crop residues) to sustain fertility and structure.
- Practice erosion control on sloping lands through terracing or vegetative cover.
- Use crop rotation and intercropping to maintain soil health and organic matter balance.

### **Suitable Crops:**

Ideal for **tea, coffee, maize, beans, Irish potatoes, pyrethrum, and vegetables**, as well as fruits like avocados and bananas.